

6-4-1: Examining Transceivers and Connector Types

At the end of this episode, I will be able to:

1. Identify and explain transceivers and connector types, given a scenario.

Learner Objective: *Identify and explain transceivers and connector types, given a scenario.*

Description: In this episode, the learner will examine various transceiver and connection considerations. We will explore standards, transceiver form factors such as SFP and QSFP, TIA Category standards and more.

- What are common considerations when using wired cabling?
 - o What is the current setup being used (most commonly Ethernet, maybe Fiber channel for storage-area networks)
- With Ethernet being the most common, what are some considerations?
 - o On the workstation end:
 - ♣ Cables = copper vs. fiber
 - o Copper
 - ♣ Plenum vs. Non-Plenum Cable -plenum cables are coated with flame-retardant materials and are used in air circulation spaces. In contrast, non-plenum cables are used in non-air circulating spaces and are toxic when burnt, and less fire retardant.
 - ♣ Cable type
 - ♣ 802.3 - (CAT5 = 100 Mbps, CAT5e -1 Gbps, CAT6 - 1 Gbps, CAT6a - 10 Gbps, CAT7 - 10 Gbps, CAT8 - 40 Gbps)
 - ♣ Connectors
 - ♣ RJ-45
 - ♣ A common connector type used for Ethernet
 - ♣ Also known as an 8P8C connector
 - ♣ Commonly wired with TIA-568 wiring standard
 - ♣ TIA-568A
 - ♣ Green and orange wire pairs on pins 1 and 2, 3 and 6, respectively, and blue and brown wire pairs remain unchanged on pins 4 and 5, 7 and 8, respectively, in the RJ-45 connector.
 - ♣ TIA-568B
 - ♣ Orange wire pair on pins 1 and 2, and green wire pair on pins 3 and 6; blue and brown wire pairs are consistent with TIA-568A, on pins 4 and 5, 7 and 8, respectively, in the RJ-45 connector.
 - ♣ RJ-11 Connector
 - ♣ Standard interface for voice-grade telephone equipment.
 - ♣ Compact, modular design.
 - ♣ Usually has four or six positions and contacts.
 - ♣ Connects to modules by plugging into standard jacks on devices like telephones, modems, and fax machines.
- What higher bandwidth demands, like data centers?
 - o Fiber
 - ♣ Ethernet and Fiber Channel
 - ♣ Single-mode fiber - allows only one light mode to pass through, enabling longer distance transmission. The core diameter is 9 microns
 - ♣ Multimode fiber - allows multiple modes, suitable for shorter, high-bandwidth applications. Core diameter 50 & 62.5 micron

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Copper

- ♣ Direct Attach Copper (DAC) - a type of twinaxial cable that features two inner conductors for high-speed, short-range data transmission, often used in data centers.
 - o Transceiver - a device that transmits and receives data signals, allowing for the conversion and processing of these signals across various communication mediums.
 - o Form Factors -
 - ♣ Small Form-factor Pluggable (SFP) - A compact, hot-pluggable transceiver used for telecommunication and data communications applications.
 - ♣ Quad Small Form-factor Pluggable (QSFP) - A compact, hot-pluggable transceiver used to carry more bandwidth than a single SFP transceiver, commonly used in data centers.
 - What are the different cable connectors in networks?
 - o Subscriber Connector (SC) - A push-pull style connector that is widely used in fiber optic communications.
 - o Local Connector (LC) - A small form-factor fiber optic connector commonly used in high-density applications due to its small size.
 - o Straight Tip (ST) - A fiber optic connector with a twist-locking mechanism used in single and multimode networks.
 - o Multi-fiber Push On (MPO) - A high-density fiber optic connector designed for data communications, capable of connecting multiple fibers in a single connector.
 - o Registered Jack (RJ)11 - A connector type primarily used for telephone wiring, and also known as an 8P8C connectors
 - o F-type - A connector mainly used for coaxial cable and is common in television, satellite, and cable modem applications.
 - o Bayonet Neill–Concelman (BNC) - A quick connect/disconnect RF connector used for coaxial cable, especially in video and RF applications.
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- Additional Resources:
 - o If applicable